

Prenex Conjunctive Normal Form (PCNF) in Predicate Logic

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Term

Terms are defined recursively as follows:

- A variable, constant or 0-ary function symbol is a term.
- If f^n is an n -ary function symbol ($n > 0$) and $t_1, t_2, \dots, t_n$ are terms, then $f^n(t_1, t_2, \dots, t_n)$ is a term.

Table : Examples

Expressions	Meaning of Symbols	is a term
x	x is a variable	YES
"An"	"An" is a constant	YES
<i>leapyear</i> (.)	<i>leapyear</i> is a predicate	NO
<i>currentyear</i> ()	<i>currentyear</i> is an 0-ary function	YES
<i>numberofday</i> (.)	<i>numberofday</i> is a 1-ary function	YES
<i>numberofday</i> (<i>currentyear</i> ())	<i>currentyear</i> is passed an argument	YES
<i>check</i> (<i>leapyear</i> (.))	<i>check</i> is a 1-ary function; <i>leapyear</i> is passed an argument	NO

Atomic formula

An *atomic formula* p is an n -ary ($n \geq 0$) predicate followed by a list of n arguments where each argument t_i is a term: $p(t_1, t_2, \dots, t_n)$.

Notation

- 1 Functions are denoted by f, g, h possibly with subscripts; e.g., f_1, f_2 , etc.
- 2 The superscript denoting the arity of the function will not be written since the arity can be inferred from the number of arguments.

Prenex CNF

A formula is in **prenex conjunctive normal form** (PCNF) iff it is of the form:

$$Q_1 x_1 \dots Q_n x_n M$$

Where,

- 1 Q_i are quantifiers.
- 2 M is a quantifier-free formula in CNF.

Table : Examples

Formulas	is PCNF
$\forall y \forall z ([p(f(y)) \vee \neg p(g(z)) \vee q(z)] \wedge [\neg q(z) \vee \neg p(g(z)) \vee q(y)])$	YES
$\forall x \exists y ([p(x, f(y)) \vee \neg q(y)] \wedge [r(x) \vee s(g(y), x, t(x, y))])$	YES
$\forall x \forall y ([p(x, y) \vee q(x)] \wedge [s(x) \vee r(x, y)] \vee [p(x, f(y)) \wedge q(y)])$	NO
$\forall x \forall y ([\forall z p(x, y, z) \vee q(x, y)] \wedge [s(x, y) \vee r(x)])$	NO

Clausal Form

- Let A be a closed formula in PCNF whose prefix consists only of universal quantifiers.
- The clausal form of A is written as a set of clauses.

Table : Examples

Formulas
$\forall y \forall z ([p(f(y)) \vee \neg p(g(z)) \vee q(z)] \wedge [\neg q(z) \vee \neg p(g(z)) \vee q(y)])$ $\rightarrow \{\{p(f(y)), \neg p(g(z)), q(z)\}, \{\neg q(z), \neg p(g(z)), q(y)\}\}$

Skolem Algorithm

- 1 Rename bound variables so that no variable appears in two quantifiers.
- 2 Eliminate all binary Boolean operators other than $\vee$ and $\wedge$.
- 3 Push negation operators inward, collapsing double negation, until they apply to atomic formulas only.
- 4 Extract quantifiers from the matrix.
- 5 Use the distributive laws to transform the matrix into CNF.
- 6 Replace existential quantifiers by Skolem functions
- 7 Write formula in clausal form by dropping the (universal) quantifiers and writing the matrix as sets of clauses.